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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 06/30/2022 |
| **PI Name:** | Diaz Artiles, Ana Ph.D. | | |
| **Project Title:** | Predicting Acute Cardiovascular and Ocular Changes due to Changes in the Gravitational Vector and Effects of Countermeasures | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes<br>(2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | adartiles@tamu.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 617-909-0644 |
| **Organization Name:** | Texas A&M University | | |
| **PI Address 1:** | Aerospace Engineering Department | | |
| **PI Address 2:** | 701 Ross Street | | |
| **PI Web Page:** | | | |
| **City:** | College Station | **State:** | TX |
| **Zip Code:** | 77843-0001 | **Congressional District:** | 17 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B |
| **Start Date:** | 09/01/2020 | **End Date:** | 06/30/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 4 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 06/30/2025 per V. Lehman/JSC (Ed., 4/20/23)<br>NOTE: End date changed to 8/31/2023 per NSSC information (Ed., 8/22/22)<br>NOTE: End date changed to 8/31/2022 per NSSC information (Ed., 8/13/21) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Dunbar, Bonnie Ph.D. ( Texas A&M University ) | | |
| **Grant/Contract No.:** | 80NSSC20K1521 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

Exposure to weightlessness results in the removal of hydrostatic pressure gradients and a permanent headward fluid shift, causing a redistribution of blood. Additionally, in space postural changes don't occur and thus, astronauts are not exposed to daily fluid shifts (between supine and upright postures) as we are on Earth. This has currently unknown consequences, but it might be related to a series of neuro-ocular and functional changes developed in some astronauts during both short and long-duration spaceflight, collectively known as Spaceflight Associated Neuro-Ocular Syndrome (SANS). While the exact etiology of SANS is currently unknown, chronic fluid redistribution affecting intravascular, interstitial, and cerebrospinal fluids and pressures is widely hypothesized to be a contributing factor. Additionally, recently demonstrated stagnant and retrograde blood flow and venous thrombosis in the left internal jugular vein during spaceflight could also be associated with sustained headward blood and tissue fluid shift.
Based on the current knowledge and hypotheses, countermeasures focused on producing hydrostatic gradients or reducing the microgravity-induced fluid shift, such as lower body negative pressure (LBNP) or centrifugation, become particularly interesting. LBNP shifts fluids from the head to the lower body is expected to reduce the translaminar pressure across the eye and thus, might prevent some of the ocular changes and remodeling associated with SANS. LBNP is also a promising countermeasure to enhance venous flow in the upper body. Similarly, head-to-foot (Gz) centrifugation also produces a fluid shift and induces hydrostatic gradients within the vessels in the body, which may also reduce intraocular pressure (IOP), reduce the translaminar pressure, and enhance venous flow. However, at present, it is not possible to estimate the overall physiological response (neither acute nor long term) of a particular "dose" of artificial gravity (AG) or LBNP, and more specifically and relevant to our purposes, the response in the upper body and around the eye. As a first step and in the context of this short project, we propose to focus on acute responses. Our objective is to generate acute gravitational dose-response curves of cardiovascular (CV) and ocular variables due to changes in the gravitational vector, with and without countermeasures. We propose to conduct a series of experimental studies (using tilt table, LBNP, and centrifugation), and combine the results with numerical modeling to leverage the advantages of both experimental and computational methodologies. Then, we propose to exert the full flexibility of the numerical framework to investigate CV and ocular responses in additional configurations where data collection is difficult, expensive, or infeasible.

**Task Description:**

A short-description of the experiments is shown below:

- Experiment 1: Tilt table. Subjects will be exposed to multiple tilt angles, from 45° HUT (head up tilt) to 45° HDT (head down tilt), in both prone and supine configurations. CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of tilt angle.

- Experiment 2: LBNP. Subjects will be exposed to multiple levels of negative pressure (from 0 to -50 mmHg), in both supine and 15° HDT. Similarly, CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of external pressure.

- Experiment 3: Centrifugation. Subjects will be exposed to multiple levels of artificial gravity on the human short-radius NASA centrifuge that is being relocated and installed at Texas A&M University (TAMU). CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of g-level.

In addition to the experiments, we will develop a numerical model from two different but complementary lumped-parameter models that have been previously validated and published in the literature: 1) A full-body model that provides an accurate simulation of short-term CV regulatory changes during changes in the gravitational vector (i.e., different tilt angles, centrifugation) and countermeasures (i.e., LBNP, exercise). This model features a more accurate body representation, including a cardiac pacemaker (i.e., pulsatile waveform), baroreceptor and cardiopulmonary feedback control of heart and arterial parameters, and a large number of compartments, particularly in the Gz direction, allowing for an improved representation of hydrostatic gradients in this direction and overall body hemodynamics. 2) An eye model that simulates volume/pressure alterations in the eye during gravitational changes. We will combine both approaches leveraging the advantages of each one of them into a unique more powerful and versatile model to study SANS, the cranial venous system, and other fluid shifts mechanisms and the effects of potential countermeasures. The model will be validated with the experimental data generated during the experiments.

Results from this investigation will inform current and future countermeasure development and in-flight prescriptions.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

This project investigates the influence of gravity (or more generally fluid shifts) on cardiovascular and ocular responses using a variety of interventions, including tilt paradigms, Lower Body Negative Pressure (LBNP), and centrifugation. Results will provide critical information for current and future cardiovascular and ocular related countermeasures and in-flight prescriptions. In addition, this research effort has direct application to cardiovascular and ocular pathologies on Earth, for example, orthostatic hypotension and conditions related to cardiovascular regulation.

The objective of this ground-based research effort is to generate acute gravitational dose-response curves of cardiovascular and ocular variables when exposed to different types of orthostatic stress. We propose to use both experimental and computational approaches to leverage the advantages of each one of these research methodologies. Our study consists of three different ground experiments where the same 12 male subjects are exposed to different levels of tilt (from 45° HUT to 45° HDT, experiment 1), different levels of lower body negative pressure or LBNP (from 0 mmHg to -50 mmHg, in both supine and -15° HDT, experiment 2), and centrifugation (from 0g to 2g, measured at the center of mass, experiment 3). In all experiments, we are collecting objective measures related to human performance and cardiovascular regulation, including continuous CV variables (Finapres NOVA), non-invasive cardiac output (Innocor, Innovision), electrocardiogram (ECG) and autonomic responses (Finapres NOVA), intraocular pressure (IOP) (I-care tonometer), brachial blood pressure for calibration (Omron), metabolic data (Innocor), blood volume (Blood Tec), Internal Jugular Vein (IJV) pressure (Vein Press), and cross-sectional areas and flows of the Internal Jugular Vein or IJV (left and right side) and the Common Carotid Arteries (CCA, left and right side) (Ultra sound VScan Extend and Butterfly iQ+). We will derive other metrics such as Ocular Perfusion Pressure. When possible, measurements are continuously monitored during the pre- (baseline), during-, and post-experimental testing (e.g., Finapres); otherwise, measurements are collected after an appropriate time of exposure to the intervention to ensure steady state (typically 5 min). During baseline, all data are also collected in upright seated position. This year we received an additional augmentation grant in order to extend our study from male subjects to also include female subjects (tilt and LBNP) and capture sex-differences in the gravitational dose-response curves.

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| **Task Progress:** | To date, we have successfully completed the first two experiments (i.e., tilt and LBNP) for male subjects. It was necessary to recollect the data from the LBNP experiment (originally collected in year 1) due to a hardware issue with the LBNP chamber identified in early 2022. Data collection is currently in progress for female subjects. Data analysis has been completed for experiment 1 (male subjects) and the results have been captured in the following peer reviewed publications:<br><br>• Whittle R.S. and Diaz-Artiles A. “Gravitational effects on jugular and carotid hemodynamics in graded head up and head down tilt”. [in review].<br><br>• Whittle R.S., Keller N., Hall E.A., Vellore H.S., Stapleton L.M., Findlay K.H., Dunbar B.J., and Diaz-Artiles A. “Gravitational dose response curves for acute cardiovascular hemodynamics in a tilt paradigm”. Journal of the American Heart Association 2022, [in press]. doi: 10.1161/JAHA.121.024175<br><br>Data analysis of the LBNP experiment is currently underway (for male and female subjects in parallel). For experiment 3, we are planning to use the Aerospace Human Centrifuge at TAMU, which is a NASA-funded facility that was formerly at the University of Texas Medical Branch (UTMB) in Galveston (TX). The final installation and check-out of the centrifuge are currently underway. This process has experienced further delays since year 1, but we expect to have the centrifuge operational later this year.<br><br>Further, during the course of our data collection for experiments 1 and 2, we published a manuscript covering the differences observed in measurement of cardiac output in altered gravity conditions using inert gas rebreathing and pulse contour analysis. These data are captured in the following peer-reviewed publication:<br><br>• Whittle R.S., Stapleton L.M., Petersen L.G., and Diaz-Artiles A. “Indirect measurement of absolute cardiac output during exercise in simulated altered gravity is highly dependent on the method”. Journal of Clinical Monitoring and Computing 2021, 1-12. doi: 10.1007/s10877-021-00769-y<br><br>We are also investigating gravitational dose responses using modeling approaches. We are developing a comprehensive numerical framework capable of predicting the expected acute cardiovascular responses and ocular changes when exposed to different types of orthostatic stress. We will use our own experimental data to validate the model.<br><br>To date, we modified an existing numerical model of the eye (Nelson et al., 2017) to model Intraocular Pressure (IOP) during a full range of tilt positions (360°, which covers both prone and supine positions). Experimental data collected in 13 subjects supported the hypothesis that IOP is statistically significantly higher in prone position than in supine position due to the extra hydrostatic column between the eye globe and the coronal plane, and this is true at most of the tilt angles investigated. Our modified version of the model successfully reproduced these results; the results are captured in the following peer-reviewed publication:<br><br>• Petersen L.G., Whittle R.S., Lee J.H., Sieker J., Carlson J., Finke C., Shelton C.M., Petersen J.C.G., and Diaz-Artiles A. “Gravitational effects on intraocular pressure and ocular perfusion pressure”. Journal of Applied Physiology 2022, 132: 24-35. doi: 10.1152/japplphysiol.00546.2021<br><br>We have extended our full body model to include a detailed simulation of the head incorporating multiple cranial venous drainage pathways. We are currently validating the results of this addition to the full body model, and anticipate incorporating our eye model into the full body simulation later this year. This complete model will be validated with data from all three experiments (i.e., tilt, LBNP, and short radius centrifugation). This will leverage the advantages of each one of them into a unique more powerful and versatile model to study SANS, the cranial venous system, and other fluid shifts mechanisms and the effects of potential countermeasures.<br><br>Additionally, we have conducted a comprehensive sensitive analysis in a “full body” numerical model (Diaz Artiles et al., 2019) to investigate individual differences in cardiovascular responses to orthostatic stress (Whittle and Diaz-Artiles 2021). Conditions simulated include constant gravity conditions (from 0g to 1g, in increments of 0.25g), and gravitation stress generated by a short-radius centrifuge, which induce a strong gravity gradient in the head-to-toe direction (from 0g to 1g, in increments of 0.25g, measured at the center of mass). Results from the constant gravity conditions show that model parameters related to the length, resistance, and compliance of the large veins and parameters related to right ventricular function have the most influence on model outcomes. For most outcome measures considered, parameters related to the heart are dominant. Results highlight which model parameters to accurately value in simulations of individual subjects’ CV response to gravitational stress, improving the accuracy of predictions. Influential parameters remain largely similar across gravity levels, highlighting that accurate model fitting in 1 g can increase the accuracy of predictive responses in reduced gravity. These research efforts have been captured in the following peer-reviewed publication:<br><br>• Whittle R.S. and Diaz-Artiles A. “Modeling individual differences in cardiovascular response to gravitational stress using a sensitivity analysis”. Journal of Applied Physiology 2021, 130: 1983-2001<br><br>Similar analysis is currently being conducted in the centrifugation conditions and a manuscript is in preparation. |
| **Bibliography Type:** | Description: (Last Updated: 07/10/2026) |
| **Abstracts for Journals and Proceedings** | Real Fraxedas F, Whittle RS, Diaz-Artiles A. "Modeling gravitational dose-response curves during tilt, LBNP, and centrifugation." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022 |
| **Abstracts for Journals and Proceedings** | Diaz-Artiles A, Whittle RS, Real Fraxedas F, Hall EA, Vellore HS, Dunbar BJ. "Gravitational dose-response curves during tilt, LBNP, and centrifugation." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022.<br>Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022 |
| **Abstracts for Journals and Proceedings** | Whittle RS, Keller N, Hall EA, Vellore HS, Stapleton LM, Findlay KH, Dunbar BJ, Diaz-Artiles A. "Hemodynamic and autonomic response of the cardiovascular system to tilt: gravitational dose-response curves." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022.<br>Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022 |

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| **Abstracts for Journals and Proceedings** | Whittle RS. "Quantifying and modeling the acute cardiovascular response to altered-gravity and spaceflight countermeasures." 2022 Rising Stars in Aerospace Symposium, Boulder, CO, May 12-13, 2022. Abstracts. 2022 Rising Stars in Aerospace Symposium, Boulder, CO, May 12-13, 2022. , May-2022 |
| **Abstracts for Journals and Proceedings** | Whittle RS, Vellore HS, Hall EA, Real Fraxedas F, Findlay KH, Stapleton LM, Dunbar BJ, Diaz-Artiles A. "Cardiovascular, autonomic, and cephalad dose-response to graded lower body negative pressure." 2022 Aerospace Medical Association (AsMA) Annual Scientific Meeting 2022 (Aerospace Medical Association/Undersea & Hyperbaric Medical Society), Reno, NV, May 22-26, 2022. Proceedings. 2022 Aerospace Medical Association (AsMA) Annual Scientific Meeting 2022 (Aerospace Medical Association/Undersea & Hyperbaric Medical Society), Reno, NV, May 22-26, 2022. , May-2022 |
| **Articles in Peer-reviewed Journals** | Whittle RS, Stapleton LM, Petersen LG, Diaz-Artiles A. "Indirect measurement of absolute cardiac output during exercise in simulated altered gravity is highly dependent on the method." J Clin Monit Comput. 2021 Oct 22:1-12. https://doi.org/10.1007/s10877-021-00769-y ; PMID: 34677821 , Oct-2021 |
| **Articles in Peer-reviewed Journals** | Petersen LG, Whittle RS, Lee JH, Sieker J, Carlson J, Finke C, Shelton CM, Petersen JCG, Diaz-Artiles A. "Gravitational effects on intraocular pressure and ocular perfusion pressure." J Appl Physiol. 2021 Dec 30;132(1):24-35. https://doi.org/10.1152/japplphysiol.00546.2021 ; PMID: 347625251 , Dec-2021 |
| **Articles in Peer-reviewed Journals** | Whittle RS, Keller N, Hall EA, Vellore HS, Stapleton LM, Findlay KH, Dunbar BJ, Diaz-Artiles A. "Gravitational dose-response curves for acute cardiovascular hemodynamics in a tilt paradigm." J Am Heart Assoc. 2022 Jul 19;11(14):e024175. https://doi.org/10.1161/JAHA.121.024175 ; PMID: 35861832 , Jul-2022 |